

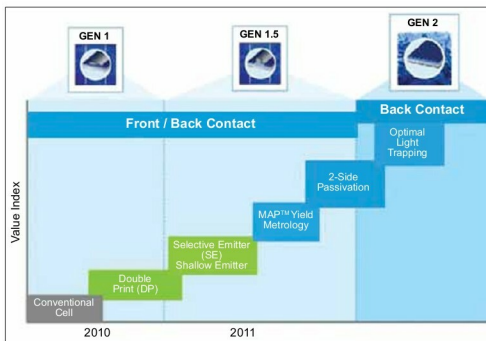


Fig. 1: Crystalline-silicon roadmap

Companies may take for granted that all of these variables are known, or can be easily and closely estimated. However, this is not always the case. A realistic estimate of these variables requires inputs from process metrology and regular cost updates for direct materials and consumables. These numbers are often tied to spot market fluctuation.

The crystalline silicon (c-Si) solar cell technology roadmap (Fig. 1) clearly indicates acceleration towards differentiation; both evolutionary and revolutionary approaches are finding traction. In both approaches, the number of process steps is increasing. Consequently, the process variability and the manufacturing uncertainty associated with the most promising cell concepts are increasing too.

A different approach

It becomes clear that the TCO as it is currently calculated might not reflect the actual worth of an investment over its entire utilisation period. A tool that can differentiate itself by its ability to adapt to changing technologies, and therefore provide greater value over time, is becoming more and more important—especially in a fast-growing and still-maturing market such as photovoltaics.

In semiconductor industry, the capital portion of the cost of goods and services is typically in 30-50 per cent range. For c-Si solar cell manufacturing segment, it is typically between 5 and 20 per cent. While this is an important cost, it is less so c-Si manufacturers than to semiconductor manufacturers. The trend we see is that whenever a tool can flexibly and rapidly embrace differentiated process variants or additions, that tool will be preferred.

To drive down the cost per watt of solar energy, it is important to achieve high flexibility and overall equipment effectiveness. Equally important is to effectively and rapidly adopt new recipes and cell structures being developed in R&D labs. This goal requires solutions that minimise the risks of adoption.

Two examples not captured in TCO: Breakage and print quality

Process yield contributes significantly to overall equipment effectiveness and influences TCO. Yield gain can be reached through better alignment for critical process steps, and real-time control of out-of-spec material. The following sections use the examples of broken wafers and better print alignment to illustrate the costs and potential profit of improving yield factors.

Breakage. Ratio between the price of silicon wafers and final solar cells is about 60 per cent. This underscores how important material costs are for photovoltaic manufacturers. The solar market requires thinner and thinner wafers, with a roadmap down to 100 μm . Thinner wafers, however, break more easily and require more delicate and precise processing in a manufacturing line.

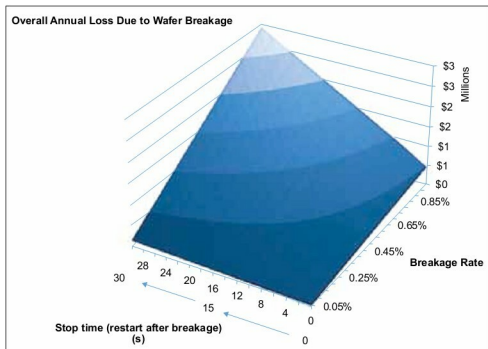


Fig. 2: Dramatic reduction in breakage provides a \$0.5 million per year margin opportunity (assuming 24 million wafers per year at \$2.10 per wafer)